

assignment _newton1

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:33pm

Quiz Instructions

1 attempt for this one.

Remember the simulation in class. NO TUTORIAL. If you are not sure ask me after class on Tuesday

$F_{net} = 0$ -> the object is at rest or moving at a constant velocity : like pushing a sofa at a constant speed since push = friction or a parachute reaching terminal speed since drag = weight.

$F_{net} = \text{constant}$ -> the object accelerates at a constant rate.

Newton's second law -> $F_{net} = \text{mass} \times \text{acceleration}$ or up-down = $m a$ or right - left = ma



Question 1 10 pts

Whenever the net force on an object is zero, its acceleration

☐

may be less than zero.

☐

is zero.

☐

may be more than zero.



Question 2 10 pts

If an apple experiences a constant net force, it will have a constant

☐

more than one of the above

☐

velocity.

☐

position.

☐

speed.

☐

acceleration.



Question 3 10 pts

A mobile phone is pulled northward by a force of 10 N and at the same time pulled southward by another force of 15 N. The resultant force on the phone is

☐

5 N.

☐

0 N.

☐

25 N.

☐

150 N.

☐

Question 4 10 pts

The force of friction on a sliding object is 10 N. The applied force needed to maintain a constant velocity is

☐

10 N.

☐

more than 10 N.

☐

less than 10 N.

☐

Question 5 10 pts

A 10-N falling object encounters 4 N of air resistance. The net force on the object is

☐

0 N.

☐

10 N.

☐

4 N.

☐

none of the above

☐

6 N.

☐

Question 6 10 pts

The force required to maintain a constant velocity for an astronaut in free space is equal to

☐

the weight of the astronaut.

☐

none of the above

☐

the mass of the astronaut.



the force required to stop the astronaut.



zero.



Question 7 10 pts

A push on a 1-kg brick accelerates it. Neglecting friction, equally accelerating a 10-kg brick requires



none of the above



one-tenth the amount of force.



just as much force.



100 times as much force.



10 times as much force.



Question 8 10 pts

A jumbo jet has a mass of 100,000 kg. The thrust for each of its four engines is 50,000 N. What is the jet's acceleration when taking off?



none of the above



2 m/s^2



1 m/s^2



4 m/s^2



0.25 m/s^2



Question 9 10 pts

When a falling object has reached its terminal velocity, its acceleration is



zero.



g .



constant.



Question 10 10 pts

A 500-N parachutist opens his chute and experiences an air resistance force of 800 N. The net force on the parachutist is then

☐

500 N upward.

☐

800 N downward.

☐

500 N downward.

☐

300 N downward.

☐

300 N upward.

Not saved

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